

Non-Calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

You have 30 minutes.

Signature

SHOW ALL WORK FOR FULL CREDIT!

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $7 + \sqrt{-4}$

b) i^2

Find the product. Write your answer in standard form. (2 points)

2. $\sqrt{-20} \cdot \sqrt{-45}$

Simplify each power of i . (1 point each)

3.

a) i^{61}

b) $\frac{1}{i^7}$

Find the sum or difference. Write your answer in standard form. (2 points)

4. $(-3 + 8i) + (5 - 2i) - (-1 + 6i)$

Find the product. Write your answer in standard form. (2 points)

5. $(6 - i)(-2 - 3i)$

Find the quotient. Write your answer in standard form. (3 points)

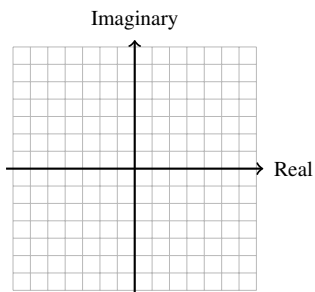
6. $\frac{3 + 11i}{1 + 2i}$

Solve the equation over the set of complex numbers. (4 points)

7. $x^2 + 29 = -4x$

Find the sum of the pair of complex numbers. Write your answer in standard form. Graph both complex numbers and their resultant. Then find the absolute value of the sum. (3 points)

8. $3 + 5i, -6 - i$



Write the complex number in rectangular form. (3 points)

9. $10(\cos 240^\circ + i \sin 240^\circ)$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $2\sqrt{3} - 2i$

b) 6

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = i\sqrt{2}$ in the Julia set? Show work.

YES / NO

ANSWER KEY – Non-Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

1.

a) $7 + \sqrt{-4}$

b) i^2

a) $7 + 2i$; $b \neq 0$.

b) $i^2 = -1$; $b = 0$.

complex, nonreal complex

complex, real

Find the product. Write your answer in standard form. (2 points)

2. $\sqrt{-20} \cdot \sqrt{-45}$

$\sqrt{-20} = 2i\sqrt{5}$, $\sqrt{-45} = 3i\sqrt{5}$ $(2i\sqrt{5})(3i\sqrt{5}) = 6i^2(5) = 30i^2 = \boxed{-30}$

Simplify each power of i . (1 point each)

3.

a) i^{61}

b) $\frac{1}{i^7}$

a) $61 \div 4 = 15 \text{ R } 1$, so $i^{61} = \boxed{i}$

b) $i^7 = i^3 = -i$; $\frac{1}{-i} \cdot \frac{i}{i} = \frac{i}{-i^2} = \boxed{i}$

Find the sum or difference. Write your answer in standard form. (2 points)

4. $(-3 + 8i) + (5 - 2i) - (-1 + 6i)$

$-3 + 8i + 5 - 2i + 1 - 6i$ real: 3 imaginary: $8 - 2 - 6 = 0$ $\boxed{3}$ (or $3 + 0i$)

Find the product. Write your answer in standard form. (2 points)

5. $(6 - i)(-2 - 3i)$

$= -12 - 18i + 2i + 3i^2 = -12 - 16i - 3 = \boxed{-15 - 16i}$

Find the quotient. Write your answer in standard form. (3 points)

6. $\frac{3 + 11i}{1 + 2i}$

$\frac{3 + 11i}{1 + 2i} \cdot \frac{1 - 2i}{1 - 2i}$ top: $3 - 6i + 11i - 22i^2 = 25 + 5i$ bottom: $1 + 4 = 5$

$\frac{25 + 5i}{5} = \boxed{5 + i}$

Solve the equation over the set of complex numbers. (4 points)

7. $x^2 + 29 = -4x$

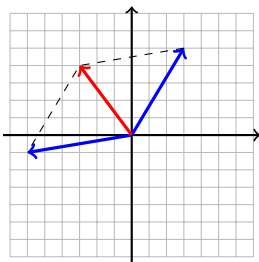
$$x^2 + 4x + 29 = 0 \quad b^2 - 4ac = 16 - 116 = -100$$

$$x = \frac{-4 \pm \sqrt{-100}}{2} = \frac{-4 \pm 10i}{2} = \boxed{-2 \pm 5i}$$

Find the sum of the pair of complex numbers. Write your answer in standard form. Graph both complex numbers and their resultant. Then find the absolute value of the sum. (3 points)

8. $3 + 5i, -6 - i$

$$(3 - 6) + (5 - 1)i = \boxed{-3 + 4i} \quad |-3 + 4i| = \sqrt{9 + 16} = \boxed{5}$$



Write the complex number in rectangular form. (3 points)

9. $10(\cos 240^\circ + i \sin 240^\circ)$

$$240^\circ: \text{QIII, reference } 60^\circ; \cos 240^\circ = -\frac{1}{2}, \sin 240^\circ = -\frac{\sqrt{3}}{2}$$

$$10\left(-\frac{1}{2}\right) + 10\left(-\frac{\sqrt{3}}{2}\right)i = \boxed{-5 - 5i\sqrt{3}}$$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

10.

a) $2\sqrt{3} - 2i$

b) 6

a) QIV. $r = \sqrt{12 + 4} = 4$
 $\tan \theta = \frac{-2}{2\sqrt{3}} = -\frac{1}{\sqrt{3}}, \text{ ref } 30^\circ$
 $\theta = 360^\circ - 30^\circ = 330^\circ$

b) positive real axis
 $r = 6, \theta = 0^\circ$

$$\boxed{4(\cos 330^\circ + i \sin 330^\circ)}$$

$$\boxed{6(\cos 0^\circ + i \sin 0^\circ)}$$

The Julia set: compute $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$. If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. (3 points)

11. Is $z = i\sqrt{2}$ in the Julia set? Show work.

$$(i\sqrt{2})^2 - 1 = 2i^2 - 1 = -2 - 1 = -3 \quad |-3| = 3 > 2$$

NO